Windows Setup Guide for Visual Studio Code & Git

🛠️ Windows Setup Guide: Visual Studio Code & Git (Intern Training)

Audience: IT Interns (Windows 10 / Windows 11)
Goal:
Interns can:

- Edit files in VS Code
- Use Git locally on Windows
- Authenticate securely to GitHub
- Follow standard enterprise workflows

Time Required: ~30–45 minutes

✅ Part 1 — Install Required Tools (Windows)

1️⃣ Install Visual Studio Code

What it is:
The standard editor used for scripts, Infrastructure as Code, and documentation.

Steps

Note: **Visual Studio Code** can also be installed from the **Company Portal**.

1. Open a browser
2. Go to:
   👉 https://code.visualstudio.com
3. Click **Download for Windows**
4. Run the installer
5. ✅ Keep **default options**
   - ✔ Add to PATH
   - ✔ Register Code as editor

✅ **Verify**

- Open **Start Menu**
- Launch **Visual Studio Code**
- You see the Welcome screen

2️⃣ Install Git for Windows

What it is:
The Git engine required for GitHub and source control.

Steps

Note: **Git** can also be installed from the **Company Portal**.

1. Go to:
   👉 https://git-scm.com/download/win
2. Download and run installer
3. Use **default settings**, but ensure:
   - ✅ **Git from the command line and 3rd-party software**
   - ✅ **Use bundled OpenSSH**
   - ✅ **Use Windows Secure Channel library**
   - ✅ **Checkout Windows-style, commit Unix-style line endings**
   - ✅ **Git Credential Manager** (VERY important)

✅ **Why these matter**

- Works with PowerShell
- Secure authentication
- Avoids line-ending issues

✅ **Verify Installation** Open **PowerShell** and run:

git --version

Expected output:

```
git version x.xx.x.windows.x
```

✅ Part 2 — One-Time Git Configuration

These settings apply to **all repos** on the intern's machine.

3️⃣ Configure Name & Email

Used for commit history and auditing.

```
git config --global user.name "First Last"

git config --global user.email "firstname.lastname@company.com"
```

✅ **Verify**

```
git config --global --list
```

4️⃣ Set Default Branch Name

Standardizes repos across teams.

```
git config --global init.defaultBranch main
```

5️⃣ Set VS Code as Default Git Editor

Avoids confusing terminal editors.

```
git config --global core.editor "code --wait"
```

✅ **Why**

- Commit messages open in VS Code
- Easier and safer for beginners

✅ Part 3 — Authenticate GitHub (Windows-Safe Method)

✅ **DO NOT use usernames/passwords**
✅ **DO NOT store PATs manually**

6️⃣ Sign in to GitHub via VS Code

Steps

1. Open **Visual Studio Code**
2. Click **Accounts icon** (bottom-left)
3. Choose **Sign in with GitHub**
4. Browser opens → authenticate
5. Approve access

✅ **What this uses**

- GitHub OAuth
- Windows Credential Manager
- MFA supported

✅ **Verify** Open VS Code terminal:

```
git pull
```

✅ No credential prompt = success

✅ Part 4 — Required VS Code Extensions

Install Extensions

Open Extensions (`Ctrl + Shift + X`) and install:

✅ Required

- **GitHub Pull Requests and Issues**
- **GitLens** (recommended)

✅ Optional (IT / Cloud)

- PowerShell
- Terraform
- Azure Tools
- YAML

✅ **Why**

- Visual diffs
- Commit history
- Easier PR reviews

✅ Part 5 — Hands-On: Clone Repo & Commit (Windows)

7️⃣ Clone Intern Training Repo

Open **VS Code Terminal (PowerShell)**

git clone https://github.com//intern-training.git

cd intern-training

code .

✅ Repo opens in VS Code

8️⃣ Create a Branch

git checkout -b feature/windows-setup-test

✅ **Rule**

> Never commit directly to `main`

9️⃣ Make a Test Change

1. Open `README.md`
2. Add:

```
Windows setup verified by <Your Name>
```

3. Save file

🔟 Commit Changes

git status

git add README.md

git commit -m "Verified Windows Git and VS Code setup"

✅ **Commit Message Rules**

- Short
- Clear
- Descriptive
- No secrets

1️⃣1️⃣ Push Branch to GitHub

git push origin feature/windows-setup-test

✅ Branch now visible in GitHub

✅ Part 6 — Pull Request (Required Workflow)

Steps

1. Open repo in GitHub
2. Click **Compare & Pull Request**
3. Add description:
   - What changed
   - Why
4. Submit PR

✅ **Explain to Interns**

- PRs = reviews
- Reviews = safety + audit trail

📝 Mini Knowledge Check (Windows)

1. Why don't we commit directly to `main`?
✅ Prevent breaking shared code

2. Where are GitHub credentials stored on Windows?
✅ Windows Credential Manager

3. What command checks file status?
✅ `git status`

✅ Windows Troubleshooting (Common Issues)

❌ Git Not Found

git : The term 'git' is not recognized

✅ Fix:

- Restart PowerShell
- Confirm Git is in PATH
- Reinstall Git if needed

❌ Credential Prompts Repeating

✅ Fix:

git config --global --unset credential.helper

Then sign in again via VS Code.

✅ Intern Setup Completion Checklist

Intern must complete **all items**:

- [ ] Windows 10/11
- [ ] VS Code installed
- [ ] Git for Windows installed
- [ ] Git configured (name/email)
- [ ] GitHub signed in via VS Code

- [ ] Repo cloned
- [ ] Branch created
- [ ] Commit created
- [ ] Pull Request submitted